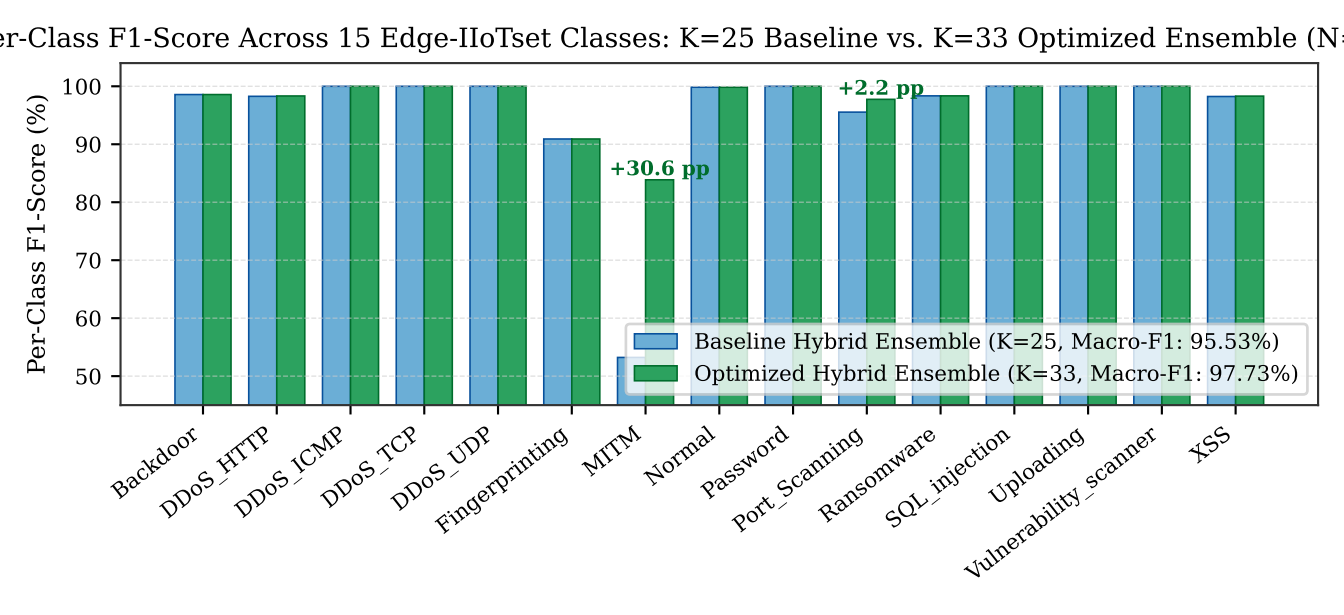
er-Class F1-Score Across 15 Edge-IIoTset Classes: K=25 Baseline vs. K=33 Optimized Ensemble (N=
Per-Class F1-Score (%)
100
90
80
70
60
50
+30.6 pp
+2.2 pp
Baseline Hybrid Ensemble (K=25, Macro-F1: 95.53%)
Optimized Hybrid Ensemble (K=33, Macro-F1: 97.73%)
Backdoor
DDoS_HTTP
DDoS_ICMP
DDoS_TCP
DDoS_UDP
Fingerprinting
MITM
Normal
Password
Port_Scanning
Ransomware
SQL_injection
Uploading
Vulnerability_scanner
XSS